NXT LVL RESEARCH / BEGINNER'S HANDBOOK

Peptides, made simple.

Understand the material. Compare the evidence.
Choose the right quantity for your research.

> **Start here**
> A peptide is a short chain of amino acids, the building blocks of proteins. Some peptides act as biological signals. A familiar name or promising study does not make every vial a medicine.

Use this guide in three steps

01. Find your research topic on page 2.
02. Compare the material profiles on pages 3-9.
03. Plan packages, quantity and preparation on pages 10-12.

Six words you will keep seeing

Word	Plain-English meaning
Vial	The small container. Its size and its contents are different things.
mg / mcg	Units of material mass. 1 mg = 1,000 mcg.
mL	Liquid volume. It does not tell you the amount of peptide.
Lyophilized	Freeze-dried. Preparation still depends on the exact material.
Blend / package	A blend combines materials in one vial. A package groups separate vials.
COA	Certificate of analysis: a test report that must match the actual lot.

Scope: 31 catalog listings grouped into 28 profiles, plus six purchasing packages. Reviewed November 3, 2026. This is a laboratory research reference, not a human treatment or injection guide. For personal health decisions, use a licensed clinician and the prescribed product's instructions. [1, 2, 4]

YOUR STARTING POINT

Find your research topic

Use the topic to find relevant reading. This chart is not a recommendation to use these products on yourself.

If your research question concerns...	Start reading about...	Page
Tissue repair and wound responses	BPC-157, TB-500 and named blends	3
Skin, inflammation and cell fuel	GHK-Cu, KPV, MOTS-c, AOD-9604	4
Appetite, body weight and energy use	Semaglutide, tirzepatide, retatrutide, SLU-PP-332	5
Growth-hormone release	Tesamorelin, sermorelin, CJC-1295 with or without DAC	6
Growth signals, sleep and anxiety	Ipamorelin, IGF-1 LR3, DSIP, Selank	7
Nervous-system or pigment signals	Semax, Melanotan-1, Melanotan-2, PT-141	8
Reproductive hormones or cell chemistry	HCG, glutathione, NAD+, preparation water	9

How to read the evidence labels

Mostly early research: mainly cells or animals. A possible effect is not a proven human benefit.
Limited human evidence: people have been studied, but the evidence does not establish a routine treatment.
Approved medicine exists*: a regulated formulation has specific medical uses. **The catalog research vial is not that medicine.**

Choose for a defined question
Check the exact identity, ingredient amounts, matching lot report, required quantity and storage instructions. A purity percentage alone does not prove that a product is free of germs or suitable for medical use. Ask about missing details before ordering.

Browse the current catalog | Review available lot reports

REPAIR SIGNALS

Tissue research and blends

The amount below is the material listed per vial, not a dose. Click a name for the current listing.

BPC-157

10 mg | Mostly early research

What it is. A small experimental peptide studied in tissue and digestive-system models.

Why it is studied. Researchers investigate repair responses in tendon, muscle and gut tissue.

Before choosing. Most evidence is from cells or animals. Reliable human healing benefits and safety are not established. [10]

TB-500

10 mg | Mostly early research

What it is. A name used for material related to thymosin beta-4, a protein involved in cell movement.

Why it is studied. Studied in tissue-repair and wound-healing research.

Before choosing. Confirm the exact molecule: the TB-500 fragment and full-length thymosin beta-4 are different. Human wound-healing benefit is unestablished. [11]

TB-500 + BPC-157

10 mg TOTAL blend | Formula details missing

What it is. A single vial containing two named materials.

Why it is studied. The combination brings together two tissue-research interests.

Before choosing. The listing does not give each ingredient's amount. Do not assume 10 mg of each, a 50/50 split, or better results together. [10, 11, 41]

GLOW

70 mg TOTAL blend | Formula details missing

What it is. A blend name, rather than one identifiable peptide.

Why it is studied. Its research purpose depends on the actual ingredients.

Before choosing. The full formula is not published in the reviewed listing. Get ingredient names and individual amounts before evaluating or ordering it. [40]

TISSUE AND METABOLISM

Skin, inflammation and cell fuel

The amount below is the material listed per vial, not a dose. Click a name for the current listing.

GHK-Cu

50 mg or 100 mg | Mostly early research

What it is. A short peptide attached to copper.

Why it is studied. Researchers study collagen production and how tissue rebuilds.

Before choosing. Cell, animal or skin-applied findings do not prove benefits from injecting it. The 100 mg vial contains twice the listed material of the 50 mg vial. [3, 12, 13]

50 mg listing / 100 mg listing

KPV

10 mg | Mostly early research

What it is. A very short peptide studied for inflammation-related signals.

Why it is studied. Relevant to laboratory questions about inflammation in tissue and gut models.

Before choosing. Human benefit and safety are not established. It is not a proven treatment for gut or skin problems. [14]

MOTS-c

10 mg | Mostly early research

What it is. A peptide linked to mitochondria, the energy-processing parts of cells.

Why it is studied. Studied for how cells handle fuel and respond to insulin.

Before choosing. Promising animal findings do not establish human energy, weight-loss or longevity benefits. [24, 25]

AOD-9604

10 mg | Human benefit unproven

What it is. A modified fragment of growth hormone, studied for fat metabolism.

Why it is studied. Investigated as a possible way to affect fat breakdown.

Before choosing. Human studies did not establish the intended weight-loss benefit. It is not a proven fat-loss or spot-reduction product. [26]

METABOLIC SIGNALS

Appetite, weight and energy

*Evidence for an approved medicine applies to its formulation, route and indicated patients, not automatically to a catalog vial.

Semaglutide

10 mg | Approved medicine exists*

What it is. Activates GLP-1, a signal involved in appetite and blood-sugar control.

Why it is studied. Approved prescription formulations have demonstrated weight-management benefits in eligible patients.

Before choosing. The research vial is not the approved medicine. Risks include severe digestive effects, pancreas and gallbladder problems, and thyroid-related restrictions. [4, 16]

Tirzepatide

10 mg | Approved medicine exists*

What it is. Activates two signals: GIP and GLP-1.

Why it is studied. Approved prescription formulations have demonstrated weight-management benefits in eligible patients.

Before choosing. A research vial is not the prescription product. Serious risks and thyroid-related restrictions apply; Zepbound labeling advises against adding another GLP-1 medicine. [4, 17]

Retatrutide

10 mg or 20 mg | Human trials; investigational

What it is. Activates three signals: GLP-1, GIP and glucagon.

Why it is studied. Clinical trials study effects on body weight and metabolism.

Before choosing. It remains investigational. Trial success does not establish the quality or safety of research vials. Digestive effects and increased heart rate are concerns. [4, 15]

10 mg listing / 20 mg listing

SLU-PP-332

10 mg | Mostly early research

What it is. An experimental small molecule. It is not a peptide.

Why it is studied. Researchers study energy use and endurance-related changes in mice.

Before choosing. Human benefit and safety are unestablished. It is not a proven exercise replacement, and a peptide-water mixing recipe cannot be assumed suitable. [38]

THE GH PATHWAY

Growth-hormone release

GH = growth hormone. IGF-1 = a growth-related signal influenced by GH. Different names can still affect the same system.

Tesamorelin

10 mg | Approved medicine exists*

What it is. Prompts the pituitary gland to release growth hormone.

Why it is studied. Approved Egrifta treats excess abdominal fat in adults with HIV-associated lipodystrophy, a specific fat-distribution condition.

Before choosing. That evidence does not make the research vial a general fat-loss product. Blood sugar, IGF-1 and fluid retention are concerns. [8, 9]

Sermorelin

10 mg | Historical medical use

What it is. A short signal that prompts growth-hormone release.

Why it is studied. Useful for understanding the body's growth-hormone signaling system; prescription products had historical growth-related uses.

Before choosing. That history does not prove adult anti-aging or recovery benefits. A research vial needs its own identity and quality documentation. [18, 20]

CJC-1295 with DAC

10 mg | Limited human evidence

What it is. A growth-hormone-releasing signal with a modification that prolongs its action.

Why it is studied. Small human studies found longer-lasting increases in growth hormone and IGF-1.

Before choosing. Higher hormone measurements are not proof of more muscle, fat loss or longevity. Clinical safety data are limited. [19, 20]

CJC-1295 without DAC

10 mg | Identity needs confirmation

What it is. A non-DAC product often marketed as modified GRF 1-29.

Why it is studied. Discussed in research on signals that release growth hormone.

Before choosing. Check the exact identity. It is different from the longer-acting DAC form, so the DAC study results cannot simply be applied to it. [19, 20]

SIGNALS AND EARLY RESEARCH

Growth, sleep and anxiety

The amount below is the material listed per vial, not a dose. Click a name for the current listing.

Ipamorelin

10 mg | Limited human evidence

What it is. Stimulates growth-hormone release through a different receptor from the CJC/sermorelin pathway.

Why it is studied. Researchers examine how that separate signaling route affects hormone release.

Before choosing. A plausible mechanism does not prove better sleep, muscle gain or recovery. Human safety information is limited and depends on the route studied. [3, 21]

IGF-1 LR3

1 mg | Mostly early research

What it is. A modified version of IGF-1, a growth-related signal.

Why it is studied. Used in experimental work on cell and tissue growth.

Before choosing. It is different from prescription mecasermin. Human muscle-building benefit is unproven; growth-related and blood-sugar effects require careful evaluation. [22, 23]

DSIP

10 mg | Limited human evidence

What it is. Short for delta sleep-inducing peptide; also called emideltide.

Why it is studied. Older studies investigated sleep and withdrawal-related outcomes.

Before choosing. The name sounds like a promise, but effectiveness for insomnia is not established. It is not a validated sleep stack ingredient. [29]

Selank

11 mg | Limited human evidence

What it is. An experimental peptide investigated for anxiety-related effects.

Why it is studied. Small human research has reported changes in anxiety symptoms.

Before choosing. Evidence is limited. Those findings do not establish a general anxiety treatment or the safety of injecting a research vial. [3, 27]

NERVOUS-SYSTEM SIGNALS

Brain, pigment and desire

*Evidence for an approved medicine applies to its formulation, route and indicated patients, not automatically to a catalog vial.

Semax

11 mg | Limited human evidence

What it is. An experimental peptide investigated for effects in the nervous system.

Why it is studied. Research has explored reduced blood flow to the brain, migraine and nerve-pain conditions.

Before choosing. Effectiveness was not established in the cited review. Most human reports involved nasal use, which does not validate injections or a focus-enhancement claim. [28]

Melanotan-1

10 mg | Approved medicine exists*

What it is. Usually refers to afamelanotide, which affects skin pigment.

Why it is studied. An approved implant improves light tolerance in adults with a rare light-sensitivity disorder called erythropoietic protoporphyria.

Before choosing. The research vial is not that implant or an established cosmetic-tanning treatment. Pigment changes need attention; sun protection remains necessary. [30]

Melanotan-2

10 mg | Limited human evidence

What it is. An experimental material affecting pigment and other melanocortin signals.

Why it is studied. A small early human study observed pigmentation and sexual effects.

Before choosing. Long-term safety is uncertain. Nausea and prolonged erections are concerns; adding PT-141 is not a validated combination. [3, 31]

PT-141

10 mg | Approved medicine exists*

What it is. Also called bremelanotide; acts on melanocortin signals involved in sexual desire.

Why it is studied. Approved Vyleesi treats a specific low-desire disorder in certain premenopausal women.

Before choosing. This is not proof of general libido or male-performance benefits. Nausea, blood-pressure changes and heart-related restrictions matter. [32]

OTHER CATALOG MATERIALS

Hormones and related materials

*Evidence for an approved medicine applies to its formulation, route and indicated patients, not automatically to a catalog vial.

HCG

5,000 IU or 10,000 IU | Approved medicine exists*

What it is. A reproductive hormone that signals the ovaries or testes.

Why it is studied. Prescription products have specific fertility and hormone-deficiency uses.

Before choosing. The research listing is not a treatment plan. Excess hormonal stimulation can be harmful; HCG is not a weight-loss aid. IU measure biological activity. [33]

5,000 IU listing / 10,000 IU listing

Glutathione

1.5 g = 1,500 mg; listed 10 mL vial | Route matters

What it is. A three-amino-acid compound involved in the cell's antioxidant defenses.

Why it is studied. Studied for oxidative stress; some oral studies measured changes in skin properties.

Before choosing. Oral findings do not prove injectable detox or recovery benefits. Contamination has caused serious reactions; vial capacity does not tell you concentration. [36, 37]

NAD+

500 mg | Limited human evidence

What it is. A helper molecule involved in cellular energy reactions. It is not a peptide.

Why it is studied. Small infusion research measured how the body processes NAD-related compounds.

Before choosing. A normal role in cells does not prove that supplementation improves energy or longevity. Injectable quality and contamination are important concerns. [34, 35]

Bacteriostatic Water

10 mL | Laboratory supply

What it is. Water with a preservative, used as a mixing liquid when the exact product or method specifies it.

Why it is studied. Its role is preparation, rather than a biological benefit.

Before choosing. It is not a universal solvent or a sterilizer. Do not automatically add it to every order; confirm compatibility and the required solvent first. [7, 39]

PACKAGES AND COMBINATIONS

What goes together - and why

A **stack** usually means using multiple products together. The website's six packages are purchasing bundles: **one vial of each listed material**, supplied separately. They are not validated human cycles. [2]

Package contents	Why the pair is discussed / what is unknown
BPC-157 10 mg + TB-500 10 mg	Both attract tissue-repair research interest. Their different proposed effects are an idea to test, not proof of faster healing together. **2 separate vials.** [10, 11]
CJC-1295 without DAC 10 mg + Ipamorelin 10 mg	They can signal GH release through different receptors. That explains the pairing idea; combined benefit and safety are not established. **2 separate vials.** [20, 21]
Tesamorelin 10 mg + Ipamorelin 10 mg	Both affect GH release. The approved tesamorelin product's results do not prove this combination works better. **2 separate vials.** [9, 21]
Sermorelin 10 mg + Ipamorelin 10 mg	Different signaling routes may be relevant to a defined GH experiment. They still overlap in the system they affect. **2 separate vials.** [18, 21]
BPC-157 10 mg + KPV 10 mg	Tissue-response and inflammation research overlap. Better human outcomes from using both have not been established. **2 separate vials.** [10, 14]
GHK-Cu 50 mg + GHK-Cu 100 mg	The same named material in two quantities: **150 mg total across 2 vials.** This is extra supply, not a complementary mechanism. [1]

A package is useful only when the study needs both materials. Compare its contents with your required quantities. One package does not mean one complete cycle.

Do not assume that semaglutide, tirzepatide and retatrutide belong together: they overlap in GLP-1 activity, and approved-product labels discourage overlapping GLP-1 regimens. Multiple GH products and PT-141/Melanotan-2 also have overlapping effects. A purchasing bundle does not establish same-vial mixing compatibility. [16, 17, 20, 31, 32]

Compare the six packages and current prices

QUANTITY PLANNER

How many vials do you need?

Start with the **total amount your documented laboratory study requires**, including repeats, controls and a justified handling allowance. Then divide by the amount in one vial.

The simple rule
Vials needed = total material required / material per vial
Always round the answer **up** to a whole vial. This is a purchasing calculation, not a dose or cycle recommendation.

Example total needed	Amount per vial	Calculation	Buy quantity
24 mg	10 mg	24 / 10 = 2.4	3 vials
44 mg	11 mg	44 / 11 = 4	4 vials
1.6 mg	1 mg	1.6 / 1 = 1.6	2 vials
120 mg	50 mg	120 / 50 = 2.4	3 vials
120 mg	100 mg	120 / 100 = 1.2	2 vials
1,200 mg	500 mg	1,200 / 500 = 2.4	3 vials
12,000 IU	5,000 IU	12,000 / 5,000 = 2.4	3 vials

All totals above are invented inventory examples. They are not suggested amounts to use. HCG calculations must stay in IU; water calculations stay in mL. Do not convert IU to mg without a validated product-specific basis.

Why a cycle length is not enough

“An eight-week cycle” does not specify a total quantity. You would also need a validated use plan, amount at each step, frequency, preparation stability and allowable waste. Most research materials here have no established human cycle. A prescriber or research protocol must supply those inputs; vial size cannot supply them.

Choose the size that fits the work

Compare usable quantity, storage limits and total cost. A larger vial may leave more unused material. Calculate each ingredient separately; a package's fixed quantities must fit the study.

For GLOW and the TB-500/BPC-157 blend, total mass is not enough to calculate each ingredient's supply. Obtain the full formula first. [40, 41]

THE BASICS

Mixing and preparation

Reconstitution means adding the specified liquid to a dry material. There is no single water volume, solvent or storage time that works for every peptide.

Understand the numbers

A vial containing **10 mg** still contains 10 mg after suitable dilution; the concentration changes. **mg/mL = material mass / final solution volume.**

Material in vial	Final volume	Concentration	Material in 0.1 mL
10 mg	1 mL	10 mg/mL	1 mg
10 mg	2 mL	5 mg/mL	0.5 mg
10 mg	3 mL	About 3.33 mg/mL	About 0.333 mg
10 mg	4 mL	2.5 mg/mL	0.25 mg
10 mg	5 mL	2 mg/mL	0.2 mg
10 mg	10 mL	1 mg/mL	0.1 mg

Arithmetic only, not mixing recommendations or administration amounts. Final volume can differ from the volume added. The solvent must be suitable and the container must hold the volume. A 10 mL container does not tell you the material's mg content.

Before you prepare anything

1. Confirm the material, lot, amount, exact solvent and written method.
2. Follow the specified laboratory setup, measuring and mixing procedure.
3. Check the expected appearance; do not improvise a fix for unexplained changes.
4. Label the concentration, date, storage conditions and validated use-by time. [5, 6, 7]

Bacteriostatic water does not sterilize contaminated powder or remove bacterial toxins. Clear liquid is not proof of suitability. Do not assume refrigeration, freezing or a universal 28-day life is appropriate. [7, 39]

View current products and prices | Lot reports | Ask about documentation

READ MORE

Sources

Numbers in the guide link to these sources. Click a title to open it. Product pages establish listed contents; studies and official labels describe their own materials, formulations and populations. Evidence reviewed November 3, 2026.

[1] NXT LVL: current catalog

[2] NXT LVL: six research packages

[3] FDA: safety concerns for selected bulk substances

[4] FDA: unapproved GLP-1 products

[5] CDC: safe preparation and injection practices

[6] Sigma-Aldrich: peptide handling guide

[7] DailyMed: bacteriostatic water label

[8] DailyMed: Egrifta SV (tesamorelin)

[9] DailyMed: Egrifta WR (tesamorelin)

[10] FDA: BPC-157 evidence review, 2026

[11] FDA: TB-500 evidence review, 2026

[12] Maquart et al.: GHK-Cu in fibroblasts

[13] Maquart et al.: GHK-Cu in rat wounds

[14] FDA: KPV evidence review, 2026

[15] Jastreboff et al.: retatrutide phase 2 trial

[16] DailyMed: Wegovy (semaglutide)

[17] DailyMed: Zepbound (tirzepatide)

[18] FDA: historical Geref (sermorelin) products

[19] Teichman et al.: CJC-1295 human study

[20] FDA: CJC-1295 evidence review

[21] Raun et al.: ipamorelin pharmacology

[22] Tomas et al.: IGF analogues in rats

READ MORE

Sources continued

Numbers in the guide link to these sources. Click a title to open it. Product pages establish listed contents; studies and official labels describe their own materials, formulations and populations. Evidence reviewed November 3, 2026.

[23] DailyMed: Increlex (mecasermin), not LR3

[24] Lee et al.: MOTS-c metabolic research

[25] FDA: MOTS-c evidence review, 2026

[26] FDA: AOD-9604 evidence review

[27] Medvedev et al.: Selank comparative study

[28] FDA: Semax evidence review, 2026

[29] FDA: DSIP/emideltide review, 2026

[30] DailyMed: Scenesse (afamelanotide implant)

[31] Dorr et al.: Melanotan-II pilot study

[32] DailyMed: Vyleesi (bremelanotide)

[33] DailyMed: Pregnyl (HCG)

[34] Grant et al.: NAD infusion pilot study

[35] FDA: ingredients for sterile compounding

[36] Weschawalit et al.: oral glutathione study

[37] FDA: glutathione contamination concerns

[38] Billon et al.: SLU-PP-332 research

[39] CDC: injection-safety observation tool

[40] NXT LVL: GLOW listing

[41] NXT LVL: TB-500/BPC-157 blend listing